Responsible AI Statement: Predicting Household Food Insecurity in Kenya

Who does this system affect?

Three groups are affected in different ways.

(a) Users: caseworkers or programme staff at an NGO or government agency who would view the model's output and decide how to act on it.

(b) Decision subjects: individual Kenyan households whose survey responses feed the model and who would be classified by it, most of whom would never know a model had been applied to their data.

(c) Voiceless affected parties: households in counties and population segments underrepresented in the Wave 4 phone survey sample. For instance, households without any phone access, who cannot appear in a phone-survey-derived dataset at all, and whose needs this model is structurally unable to represent.

What is the worst plausible outcome?

A severely food-insecure rural household is missed by the model and excluded from a prioritisation list during a resource-constrained response, because the model's recall on the severely-insecure class is measurably lower for rural households (45.9%) than urban households (58.1%). Equally concrete: a genuinely food-secure household is flagged as severely insecure, this happened to 62 of 489 truly food-secure households in the test set, nearly as many as the 64 severe cases the model correctly caught, diverting scarce aid away from those who need it most.

Who bears the cost of errors?

The cost falls almost entirely on the households being classified, not on the organisation deploying the model or the person who built it. A missed severely-insecure household bears the direct cost of continued food insecurity; a false-flagged food-secure household bears no direct harm but contributes to misallocated resources at another household's expense. This asymmetry means model errors should be monitored through their effect on the people scored, not through aggregate accuracy alone.

What redress mechanism exists?

None at present. This is a capstone research project with no operational deployment, and no appeal mechanism exists for anyone scored by it. If it were adopted operationally, we propose that any household excluded from assistance on the basis of this model's output should have the right to request a human follow-up visit or interview within a fixed window (e.g. 14 days), and that county-level, or NGO programme staff, not the model, should retain final authority over eligibility decisions.

How will the model be monitored?

The model was trained on data collected in 2020–2021 during the COVID-19 economic shock. Its patterns should not be assumed to transfer unchanged to a different kind of shock (e.g. a climate-driven flood event) or to a different time period. Before any real use, we recommend re-evaluating macro F1 and the urban/rural and county-level disaggregated tables against more recent data, and withdrawing or retraining the model if overall macro F1 falls below 0.40 or if the urban–rural severe-class recall gap widens beyond its current 12 percentage points.

Fairness Criterion Applied

We applied Equal Opportunity: among households that are truly severely food insecure, the model should identify an equal proportion regardless of urban or rural status.

We chose this over Demographic Parity because the goal is finding genuine need, not equal predicted-positive rates; and over strict Equalized Odds because false negatives (missed need) are more serious here than false positives (misdirected aid).

By this standard the model currently falls short: severe-class recall is 58.1% for urban households versus 45.9% for rural households, a gap that would need to close before this model could be considered fair for operational targeting.

Disaggregated Findings

Subgroup	N	Macro F1	Weighted F1	Severe-class recall
Urban	523	0.469	0.563	0.581

Rural	456	0.436	0.507	0.459

Full results, including county-level breakdowns, are reported in notebooks/04_evaluation.ipynb. Counties with adequate test-set size (N ≥ 20) range from macro F1 of 0.247 to 0.670, indicating substantial geographic variation not captured by the urban/rural split alone.

Datasheet Summary

This dataset represents Kenyan households reachable by phone during a national COVID-19 rapid-response survey in 2020–2021, and therefore underrepresents households without phone access or in areas with poor network coverage at the time, and cannot reflect any household dynamics that changed since collection, including effects of subsequent climate shocks this model was never trained to recognise.